

Homework2

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Question 1 : Solution:

$$\begin{aligned}\mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \\ \mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0} &\implies \mathbb{E}[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta} \\ \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n &\implies \text{Var}(\mathbf{y}) = \sigma^2 \mathbf{I}_n \\ \hat{\boldsymbol{\beta}} &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}\end{aligned}$$

$$\begin{aligned}\mathbb{E}[\mathbf{c}^T \mathbf{y}] &= \mathbf{a}^T \boldsymbol{\beta} \\ \mathbf{c}^T \mathbf{X} \boldsymbol{\beta} &= \mathbf{a}^T \boldsymbol{\beta} \quad (\forall \boldsymbol{\beta}) \\ \mathbf{X}^T \mathbf{c} &= \mathbf{a}\end{aligned}$$

Let $\mathbf{c} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} + \mathbf{d}$:

$$\begin{aligned}\mathbf{X}^T \mathbf{c} &= \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} + \mathbf{X}^T \mathbf{d} \\ \mathbf{a} &= \mathbf{a} + \mathbf{X}^T \mathbf{d} \\ \mathbf{X}^T \mathbf{d} &= \mathbf{0} \implies \mathbf{d}^T \mathbf{X} = \mathbf{0}^T\end{aligned}$$

$$\begin{aligned}\text{Var}(\mathbf{a}^T \hat{\boldsymbol{\beta}}) &= \text{Var}(\mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}) \\ &= \mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \text{Var}(\mathbf{y}) \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} \\ &= \sigma^2 \mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} \\ &= \sigma^2 \mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a}\end{aligned}$$

$$\begin{aligned}\text{Var}(\mathbf{c}^T \mathbf{y}) &= \sigma^2 \mathbf{c}^T \mathbf{c} \\ &= \sigma^2 [\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} + \mathbf{d}]^T [\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} + \mathbf{d}] \\ &= \sigma^2 [\mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} + 2\mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{d} + \mathbf{d}^T \mathbf{d}] \\ &= \sigma^2 \mathbf{a}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{a} + \sigma^2 \cdot 0 + \sigma^2 \mathbf{d}^T \mathbf{d} \\ &= \text{Var}(\mathbf{a}^T \hat{\boldsymbol{\beta}}) + \sigma^2 \|\mathbf{d}\|^2\end{aligned}$$

$$\text{Var}(\mathbf{a}^T \hat{\boldsymbol{\beta}}) \leq \text{Var}(\mathbf{c}^T \mathbf{y})$$

Question 2 : Solution:

$$\mathbf{X} = [\mathbf{x}_0 \quad \mathbf{x}_1 \quad \dots \quad \mathbf{x}_p]$$

The inner product $\mathbf{x}_i^T \mathbf{x}_j = 0$ for all $i \neq j$.

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} \mathbf{x}_0^T \mathbf{x}_0 & 0 & \dots & 0 \\ 0 & \mathbf{x}_1^T \mathbf{x}_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathbf{x}_p^T \mathbf{x}_p \end{bmatrix}$$

The inverse of a diagonal matrix:

$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} \frac{1}{\mathbf{x}_0^T \mathbf{x}_0} & 0 & \dots & 0 \\ 0 & \frac{1}{\mathbf{x}_1^T \mathbf{x}_1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\mathbf{x}_p^T \mathbf{x}_p} \end{bmatrix}$$

$$\mathbf{X}^T \mathbf{y} = \begin{bmatrix} \mathbf{x}_0^T \mathbf{y} \\ \mathbf{x}_1^T \mathbf{y} \\ \vdots \\ \mathbf{x}_p^T \mathbf{y} \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$

Multiplying the diagonal matrix by the vector yields:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = \begin{bmatrix} \frac{\mathbf{x}_0^T \mathbf{y}}{\mathbf{x}_0^T \mathbf{x}_0} \\ \frac{\mathbf{x}_1^T \mathbf{y}}{\mathbf{x}_1^T \mathbf{x}_1} \\ \vdots \\ \frac{\mathbf{x}_p^T \mathbf{y}}{\mathbf{x}_p^T \mathbf{x}_p} \end{bmatrix}$$

$$\hat{\beta}_j = \frac{\mathbf{x}_j^T \mathbf{y}}{\mathbf{x}_j^T \mathbf{x}_j} = \frac{\mathbf{x}_j^T \mathbf{y}}{\|\mathbf{x}_j\|^2}$$

Question 3 : Solution:

(a) $\mathbf{X}^T \mathbf{X} \beta_{\text{mns}} = \mathbf{X}^T \mathbf{y}$

$$\begin{aligned} \text{LHS} &= (\mathbf{V} \Sigma \mathbf{U}^T \mathbf{U} \Sigma \mathbf{V}^T) (\mathbf{V} \Sigma^{-1} \mathbf{U}^T \mathbf{y}) \\ &= \mathbf{V} \Sigma^2 (\mathbf{V}^T \mathbf{V}) \Sigma^{-1} \mathbf{U}^T \mathbf{y} \\ &= \mathbf{V} \Sigma \mathbf{U}^T \mathbf{y} = \mathbf{X}^T \mathbf{y} = \text{RHS} \end{aligned}$$

(b) $\|\beta\| \geq \|\beta_{\text{mns}}\|^2$

Let $\beta = \beta_{\text{mns}} + \mathbf{b}$, then $\mathbf{X}^T \mathbf{X} \mathbf{b} = \mathbf{0}$.

$$\mathbf{b}^T \mathbf{X}^T \mathbf{X} \mathbf{b} = 0 \implies \|\mathbf{X} \mathbf{b}\|^2 = 0 \implies \mathbf{X} \mathbf{b} = \mathbf{0}$$

$$\mathbf{U} \Sigma \mathbf{V}^T \mathbf{b} = \mathbf{0} \implies \mathbf{V}^T \mathbf{b} = \mathbf{0}$$

Orthogonality:

$$\beta_{\text{mns}}^T \mathbf{b} = (\mathbf{V} \Sigma^{-1} \mathbf{U}^T \mathbf{y})^T \mathbf{b} = \mathbf{y}^T \mathbf{U} \Sigma^{-1} (\mathbf{V}^T \mathbf{b}) = 0$$

Pythagorean theorem:

$$\|\beta\|^2 = \|\beta_{\text{mns}}\|^2 + \|\mathbf{b}\|^2 \geq \|\beta_{\text{mns}}\|^2$$

(c) Pseudo-inverse Verification

Using the Penrose properties.

(a) $\mathbf{X} \mathbf{A} \mathbf{X} = \mathbf{U} \Sigma \mathbf{V}^T \mathbf{V} \Sigma^{-1} \mathbf{U}^T \mathbf{U} \Sigma \mathbf{V}^T = \mathbf{U} \Sigma \mathbf{V}^T = \mathbf{X}$

(b) $\mathbf{A} \mathbf{X} \mathbf{A} = \mathbf{V} \Sigma^{-1} \mathbf{U}^T \mathbf{U} \Sigma \mathbf{V}^T \mathbf{V} \Sigma^{-1} \mathbf{U}^T = \mathbf{V} \Sigma^{-1} \mathbf{U}^T = \mathbf{A}$

(c) $(\mathbf{X} \mathbf{A})^T = (\mathbf{U} \Sigma \mathbf{V}^T \mathbf{V} \Sigma^{-1} \mathbf{U}^T)^T = (\mathbf{U} \mathbf{U}^T)^T = \mathbf{U} \mathbf{U}^T = \mathbf{X} \mathbf{A}$

(d) $(\mathbf{A} \mathbf{X})^T = (\mathbf{V} \Sigma^{-1} \mathbf{U}^T \mathbf{U} \Sigma \mathbf{V}^T)^T = (\mathbf{V} \mathbf{V}^T)^T = \mathbf{V} \mathbf{V}^T = \mathbf{A} \mathbf{X}$

It is the pseudo-inverse.

Question 4 : Solution:

a) Download

Download the CCpp data.

b) Data

i) The data set has **9568 rows** and **5 columns**.

Predictors: Temperature (AT), Exhaust Vacuum (V), Ambient Pressure (AP) and Relative Humidity (RH)

Trying to predict: PE (the net hourly electrical output).

ii) In the matrix Fig. 1, AT shows the strongest relationship with PE, V a weaker negative one, AP and RH almost none.

AT and V are strongly correlated.

iii) Table 1 shows the summary details.

c) Simple LR

A single-predictor regression $PE \sim X$ is fit for each variable (Table 2). All four slopes are highly significant ($p \approx 0$) but explain very different amounts of variation: AT alone accounts for about 90%, V about 76%, and AP and RH very little, giving the order $AT > V > AP > RH$.

Using Cook's distance ($4/n$). A lot of points get flagged in each fit, but this is pretty much just a sample-size thing which toss them out, refit, and the slope and R^2 hardly budge. Nothing really changes, so everything was kept. Fig. 2 shows the details.

(d) Multiple LR(linear regression)

Raising R^2 to 0.929 (Table 3).

Every coefficient is significant by an huge margin (the largest p -value among them is around 10^{-11}).

Then, $H_0 : \beta_j = 0$ was rejected for all four predictors.

(e) Univariate vs. multiple coefficients

If the single-predictor and joint estimates agreed, the four points in Fig. 3 would sit on the 45° line.

If they don't, AP drops from +1.49 to near zero and RH flips sign (+0.46 to -0.16).

(f) Nonlinear association

To test for curvature, quadratic and cubic terms one predictor were added at a time, which fitting $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3$ and the p -values of the higher-order terms (Table 4). AT, AP and RH all show clear nonlinearity. Considering V's quadratic term is far from significant ($p \approx 0.77$) and the cubic only margin.

So V is essentially linear in PE.

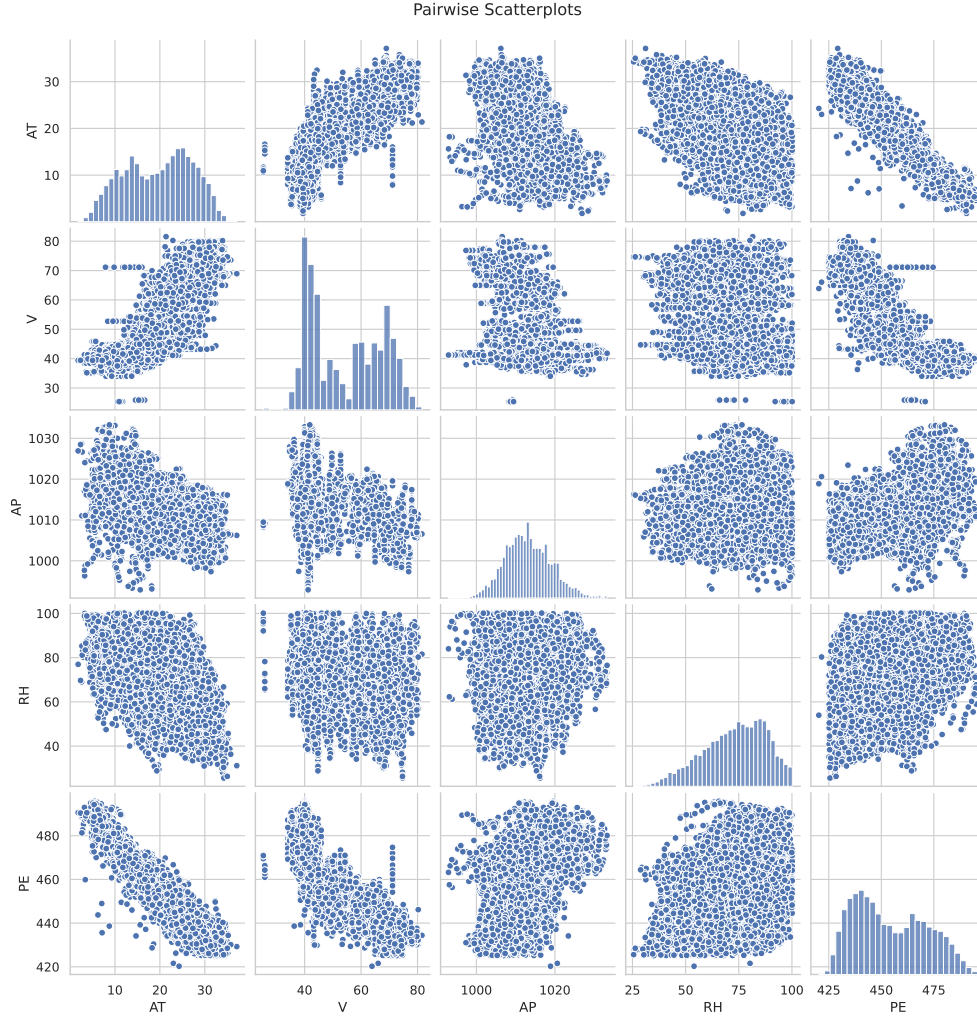


Figure 1: Pairwise scatterplots of all variables.

Table 1: Summary

	Mean	Median	Range	Q1	Q3	IQR
AT	19.651	20.345	35.30	13.510	25.720	12.210
V	54.306	52.080	56.20	41.740	66.540	24.800
AP	1013.259	1012.940	40.41	1009.100	1017.260	8.160
RH	73.309	74.975	74.60	63.328	84.830	21.502
PE	454.365	451.550	75.50	439.750	468.430	28.680

Table 2: Simple linear regression and influence diagnostics

Predictor	Slope	p -value	R^2	# influ. (%)	Slope (rm)	R^2 (rm)
AT	-2.1713	$< 10^{-300}$	0.8989	416 (4.3%)	-2.2336	0.9196
V	-1.1681	$< 10^{-300}$	0.7565	423 (4.4%)	-1.1666	0.7980
AP	1.4899	$< 10^{-300}$	0.2688	300 (3.1%)	1.7648	0.3610
RH	0.4557	$< 10^{-300}$	0.1519	249 (2.6%)	0.5252	0.1964

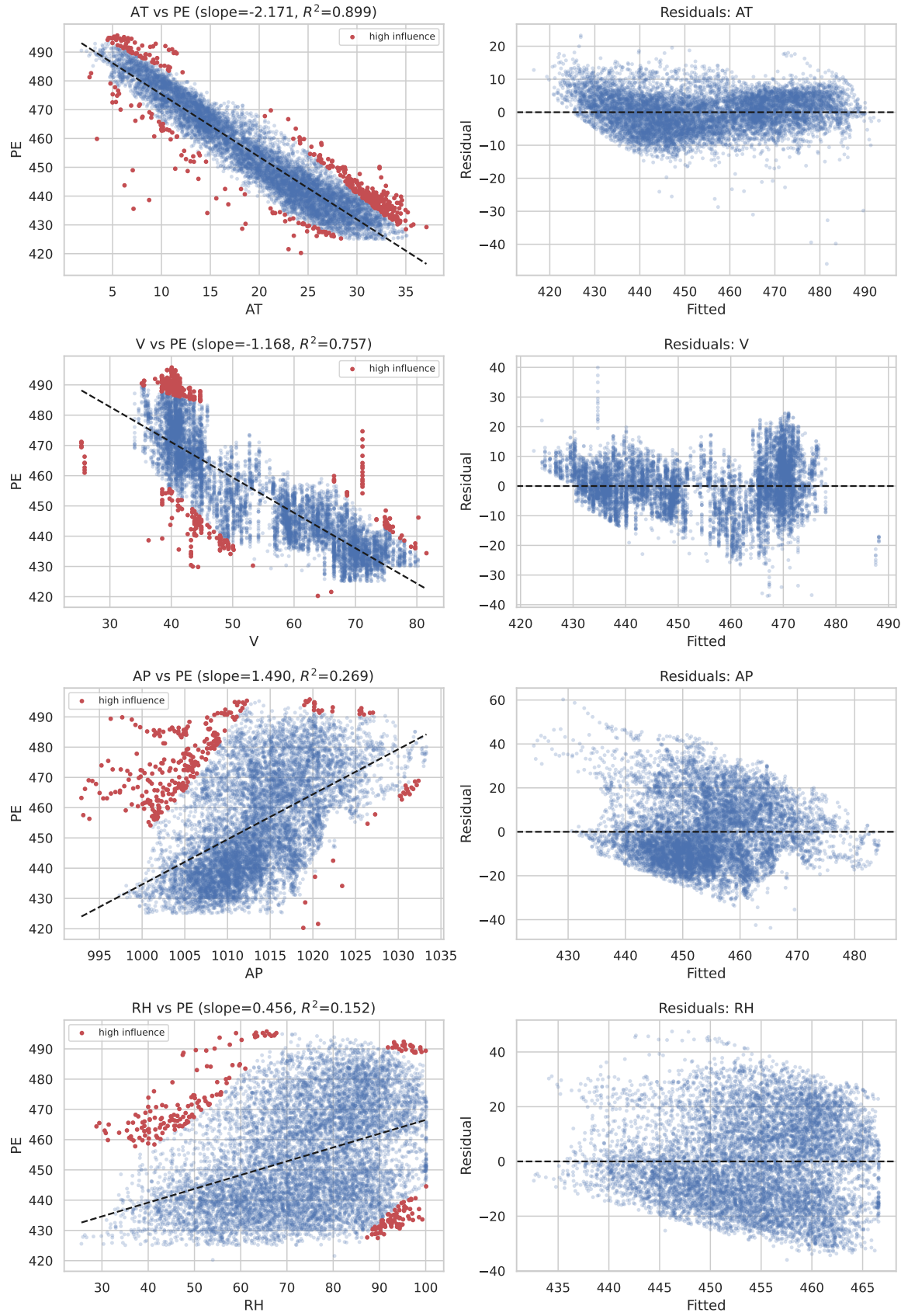


Figure 2: For each predictor: the fitted line with high-influence points marked (left), and residuals versus fitted values (right).

Table 3: Multiple linear regression (intercept = 454.609).

Predictor	Coef	Std. err	t	p -value
AT	-1.9775	0.0153	-129.34	$< 10^{-300}$
V	-0.2339	0.0073	-32.12	4.4×10^{-215}
AP	0.0621	0.0095	6.56	5.5×10^{-11}
RH	-0.1581	0.0042	-37.92	3.1×10^{-293}

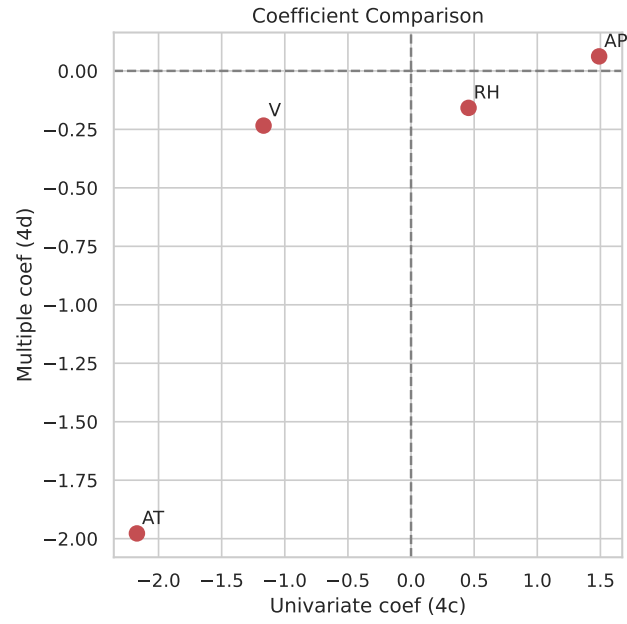


Figure 3: Single-predictor coefficients (4c) against the multiple-regression coefficients (4d).

Table 4: p -values of the higher-order terms in the cubic fits.

Predictor	X^2 p -value	X^3 p -value
AT	8.8×10^{-73}	3.7×10^{-110}
V	7.7×10^{-1}	1.4×10^{-2}
AP	3.7×10^{-17}	8.3×10^{-18}
RH	9.4×10^{-6}	1.4×10^{-5}

g) Interaction

Adding every pairwise product of the R^2 up to 0.936 (Table 5).

So far, the strongest interaction is $AT \cdot V$. Also, $AT \cdot RH$, $V \cdot AP$ and $AP \cdot RH$ are also coming out significant as well.

Only $AT \cdot AP$ and $V \cdot RH$ fail to reach significance.

h) Feature selection

On a fixed 70/30 split (`random.state=42`), the plain four-predictor model is compared against a degree-2 model with all interactions $X_i X_j$ and quadratics X_j^2 , pruned by p -value.

Backward elimination removes one term at a time and won't drop a main effect while anything built on it survives.

What's left is

$$\{ AT, V, AP, RH, AT^2, AT \cdot V, AT \cdot AP, AT \cdot RH, AP^2, AP \cdot RH, RH^2 \},$$

with V kept.

The extra terms help on the data (Table 6).

i) KNN regression

Sweeping k from 1 to 100 and tracking train and test error, once on the raw features and once after standardizing them (Fig. 4). The best test error lands at $k = 5$ (MSE 15.73) for the raw features and at $k = 4$ (MSE 14.31) for the standardized ones.

(j) Putting it together

Among every model (Table 7), the standardized KNN comes out on top, ahead of even the improved linear model.

Reasons that may explain why:

Clear nonlinear/interaction structure (parts f-g) plus $\sim 10,000$ points means local averaging captures it without assuming a functional form.

Scales differ significantly, so raw distances are dominated by AP and neighbourhoods become meaningless.

The linear model is interpretable and cheap. KNN is more accurate here but stores all training data and explains nothing.

Table 5: Full pairwise-interaction model.

Term	Coef	p -value
AT	-4.3470	6.7×10^{-2}
V	-7.6749	1.4×10^{-8}
AP	-0.1524	4.7×10^{-2}
RH	1.5709	4.2×10^{-2}
AT·V	0.0210	3.3×10^{-117}
AT·AP	0.0018	4.5×10^{-1}
AT·RH	-0.0052	1.2×10^{-10}
V·AP	0.0068	2.9×10^{-7}
V·RH	0.0008	8.6×10^{-2}
AP·RH	-0.0016	3.4×10^{-2}

Table 6: Train and test MSE on the 70/30 split.

Model	Train MSE	Test MSE
Base multiple linear regression	20.5808	21.2399
Improved (selected interactions + quadratics)	17.8908	18.6600

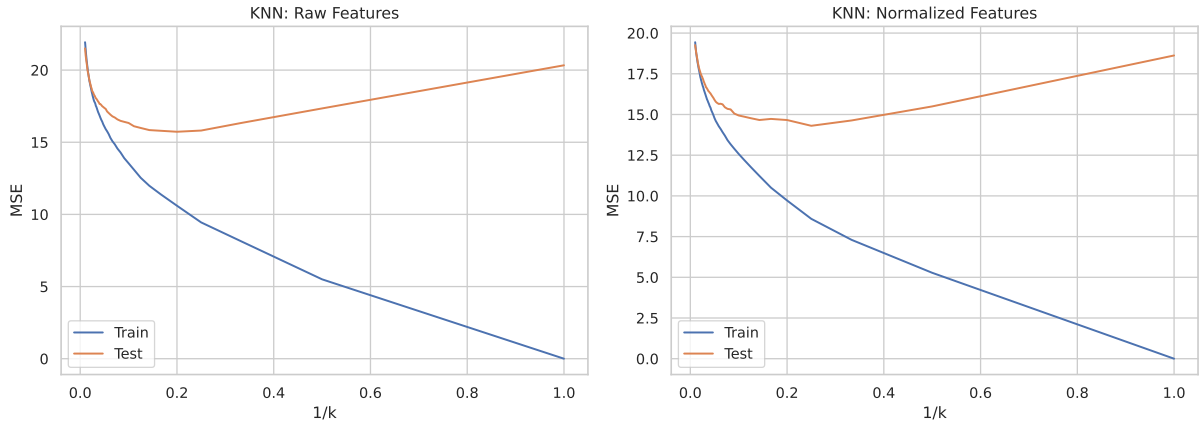


Figure 4: KNN train and test MSE against $1/k$; model complexity grows toward the right.

Table 7: Final test-MSE comparison.

Model	Test MSE
Base multiple linear regression	21.2399
Improved polynomial regression	18.6600
KNN, raw features ($k = 5$)	15.7268
KNN, normalized features ($k = 4$)	14.3057